PNPInverse

Project Overview & Motivation

Onboarding doc 00 of 06

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What this document is for

A bird's-eye view of the project: the scientific question we are trying to answer, what kind of code we write to answer it, where you fit in, and what the rest of the onboarding documents will cover. You should be able to read this in 15–20 minutes. The next docs go deep into the chemistry, the math, and the code.

Contents

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1 The big picture in one paragraph

We are building a computational model of an **electrochemical cell** — specifically the surface where an electrode meets a salt-water electrolyte and the **oxygen reduction reaction (ORR)** happens. The ORR is how oxygen gas, O_2, gets reduced to either water (H_2O) or hydrogen peroxide (H_2O_2). Which product you get, and at what voltage, depends on a tangled mix of catalyst surface chemistry, local pH, and how ions arrange themselves in the few nanometers nearest the electrode. Our group at Northwestern (the Mangan and Seitz labs) has experimental data showing that the choice of **spectator cation** — the "inert" positive ion in the salt, like K^+, Cs^+, Na^+, Li^+ — strongly changes which ORR product the cell makes, even though the cation itself doesn't appear in the chemical equation. We want to build a forward simulator that reproduces this cation-dependent behavior from physical principles, and then (later) run that simulator backward to extract the underlying kinetic constants.

2 Why this matters scientifically

- **H_2O_2 is a useful product.** Hydrogen peroxide is a green oxidant used in water treatment, paper bleaching, and potentially as a fuel. Today industrial H_2O_2 is made by the energy-intensive anthraquinone process. Electrochemical H_2O_2 generation from O_2 + electricity is a candidate replacement, but only if you can hit it *selectively* — if your electrode reduces O_2 all the way to water you get nothing useful.

- **Cation effects are mysterious and reproducible.** For decades, electrochemists have noticed that swapping $Li^+ \leftrightarrow Cs^+$ in an ORR cell can shift current densities and selectivities by factors of 2–10. Modern theories (Co & Zhang 2019, Singh 2016, Ringe 2019) blame this on cations modulating the electric double layer and the local pH near the electrode, not on cations entering the reaction. There is no first-principles forward model that nails this quantitatively.

- **Inverse parameter inference is the long game.** The rate constants for ORR (k_0, transfer coefficient α, etc.) cannot be measured directly — they are inferred by fitting an assumed kinetic model to current-voltage curves. Most groups fit toy models (no spatial transport, no double layer). Ours is one of very few attempts to fit a full PDE-based model with adjoint gradients, which is what gives the inverse problem any hope of being identifiable. **This pipeline is currently paused** until the forward model is mature enough to be trusted; that's what we are working on now.

3 What kind of code we write

3.1 The *forward* problem

Given:

- a 1D or 2D slice of the electrode–electrolyte region (typically $\sim$10–100 μm thick),
- bulk concentrations of O_2, H^+, OH^-, K^+, SO_4^{2-},
- a set of **kinetic parameters** $\{k_0^{(2e)}, k_0^{(4e)}, \alpha^{(2e)}, \alpha^{(4e)}\}$ for the two parallel ORR reactions,
- a value of the applied electrode potential V_{RHE},

the forward solver computes the steady-state distribution of every species concentration and the electric potential everywhere in the domain, and from those it computes the **disk current density** (total current at the electrode) and the **peroxide current density** (the slice of that current going into H_2O_2).

The math underneath is the **Poisson–Nernst–Planck (PNP)** system coupled to a **Butler–Volmer (BV)** boundary condition at the electrode. Doc 02 derives both. Doc 03 explains how we discretize and solve them.

3.2 The *inverse* problem (paused)

Given experimental current-voltage data from a real electrochemistry experiment, find the kinetic parameters that, when fed into the forward solver, reproduce the data. This is a nonlinear PDE-constrained optimization problem; we solve it with adjoint gradients (each forward solve is expensive, so we cannot afford finite-difference Jacobians). The math is interesting but the pipeline is paused until the forward model is trustworthy enough that fitting it is meaningful. You will not touch this in your first few months.

4 The experimental anchor: Seitz / Mangan / Ruggiero 2022

Everything we model is grounded in real data from our collaborators. The canonical paper is:

> Ruggiero, Sanroman Gutierrez, George, Mangan, Notestein, Seitz. *Probing the Relationship Between Bulk and Local Environments to Understand Impacts on Electrocatalytic Oxygen Reduction Reaction.* **Journal of Catalysis**, 2022.

Niall Mangan (one of the co-authors) is the PI of our lab on the applied-math side. Linsey Seitz's group does the experiments. The paper measures ORR currents on a CMK-3 mesoporous-carbon electrode in K_2SO_4 electrolyte at bulk pH 4–6, using a **rotating ring–disk electrode (RRDE)**. The disk current tells you how fast oxygen is being consumed; the ring current is calibrated to tell you how much H_2O_2 leaves the disk and gets re-oxidized at the (downstream) ring. Together they let you extract selectivity (% peroxide) and effective electron number n_e.

The 2022 paper is the peer-reviewed source. A 2025 conference deck extends the dataset to four cations (Li^+, Na^+, K^+, Cs^+) at the same electrochemistry. The deck is the moving target our forward model is chasing.

You should at some point read the Ruggiero 2022 PDF (`docs/papers/Ruggiero2022_JCatal_manuscript.pdf`) and skim the doc `docs/papers/Ruggiero2022_JCatal_source_paper.md` which is our internal cheat-sheet of the paper's hard numbers (the collection efficiency $N = 0.224$, the rotation rate 1600 rpm, the electrolyte composition, etc.). Doc 01 in this onboarding series is basically a chemistry tour of the same physics.

5 Where the project is right now (May 2026)

The codebase has gone through several major versions. The current production stack is sometimes called the **three-species log-c muh stack** (don't worry about the acronyms yet — doc 03 unpacks them). It models:

- three dynamic species: O_2, H_2O_2, H^+;
- one or two analytic counter-ions (K^+, SO_4^{2-}) computed from a closed-form Boltzmann-with-steric-saturation expression;
- parallel 2-electron and 4-electron ORR pathways at their Ruggiero 2022 standard potentials ($E^\circ = 0.695$ V and 1.23 V vs. RHE respectively);
- a finite **Stern compact-layer** capacitance at the electrode surface ($C_S = 0.20$ F/m^2);
- optionally, water self-ionization ($H_2O \rightleftharpoons H^+ + OH^-$) and **cation hydrolysis at the OHP** (Singh 2016 field-dependent pK_a).

We are deep in **Phase 6β**, which is the project name for adding cation hydrolysis. The most recent landed work, as of 2026-05-11, is "Step 9 B.2" — a 14-rung sweep over hydrolysis rate $\times$ activation parameter at a single calibration voltage. Doc 04 chronicles the Phase 6 history; you don't need to memorize it but the language ("Phase 6α", "v10b", "Step 9", "Phase D") comes up constantly in commits and in the Slack-style handoff notes under `docs/handoffs/`.

6 Map of the rest of these docs

Doc 01 — Electrochemistry primer. Everything you need about electrodes, the electric double layer, Butler–Volmer kinetics, and the ORR reaction set. This is the longest doc.

Doc 02 — The PNP–BV mathematical model. How the PDEs are written and what each term means. Limited PDE background assumed.

Doc 03 — Numerics and codebase tour. Newton, continuation, Firedrake, and a guided tour of the most-edited files.

Doc 04 — Phase 6β status and open questions. Where the science is right now, what's blocked, and what kinds of contributions are useful for a new contributor.

Doc 05 — Meeting agenda & getting started. The plan for our first meeting and a hands-on setup checklist.